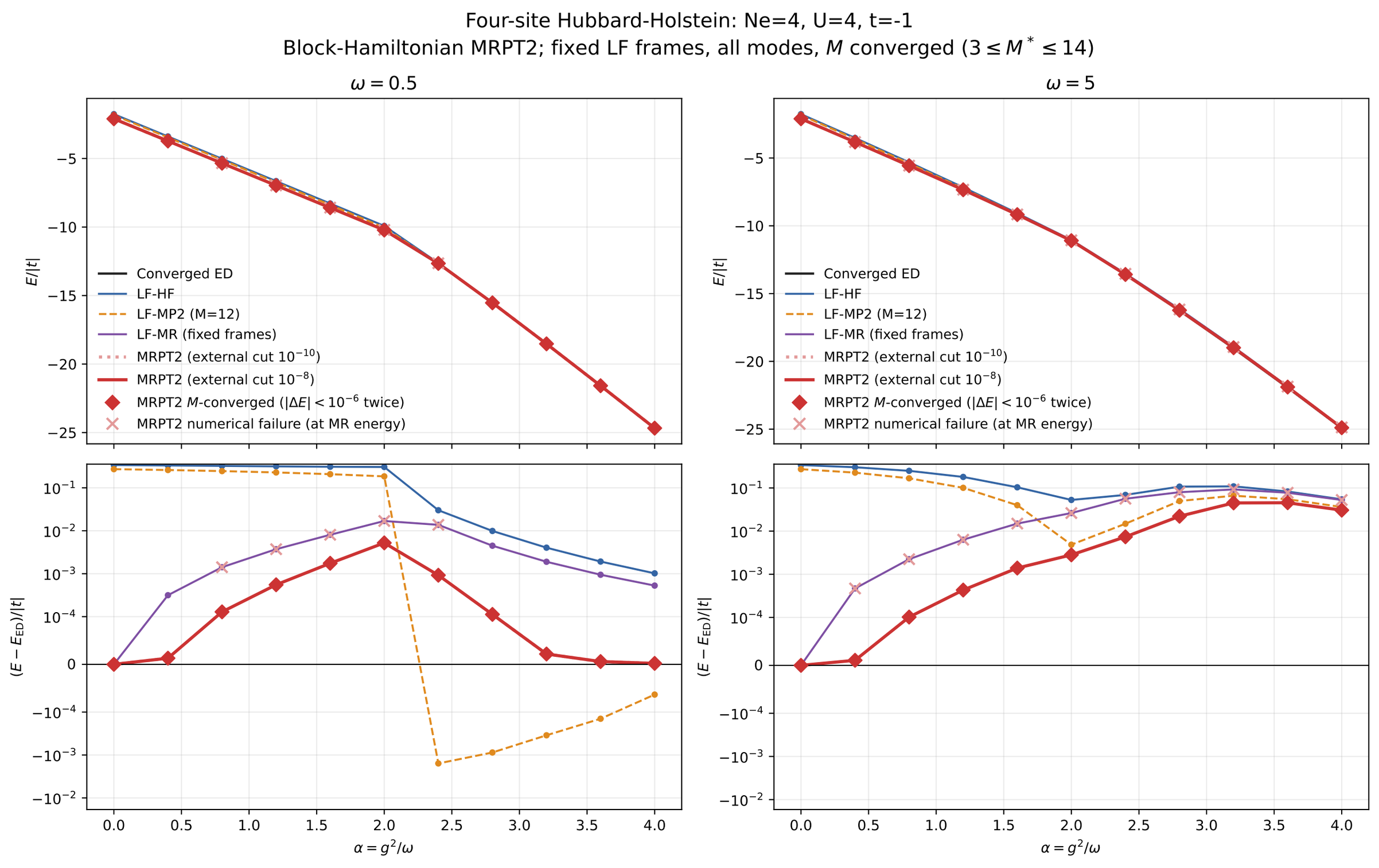
Four-site Hubbard-Holstein: Ne=4, U=4, t=-1
Block-Hamiltonian MRPT2; fixed LF frames, all modes, M converged ($3 \le M^* \le 14$)
$\omega = 0.5$
$\omega = 5$
$E/|t|$
$(E - E_{ED})/|t|$
$\alpha = g^2/\omega$
Converged ED
LF-HF
LF-MP2 (M=12)
LF-MR (fixed frames)
MRPT2 (external cut 10^{-10})
MRPT2 (external cut 10^{-8})
MRPT2 M-converged ($|\Delta E| < 10^{-6}$ twice)
MRPT2 numerical failure (at MR energy)